



Seeing the forest for the deer: Do reductions in deer-disturbance lead to forest recovery?

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ABSTRACT

High levels of deer browsing can prevent canopy tree regeneration, but little is known about changes to forest size-structure following long-term deer herd reductions. We monitored changes in forest stand structure and composition in southwestern Ontario, Canada, over 28-years using permanent plots. Our study site was the largest remaining tract of Carolinian (deciduous) forest in Canada (11 km²), a habitat type that contains up to a fifth of Canada's species at risk and is under intense anthropogenic pressures. We recorded declines in all tree size classes between 1981 and 1996, during which densities of white-tailed deer (*Odocoileus virginianus*) reached a peak of 55 deer km⁻². Despite significant and sustained deer herd reductions between 1996 and 2009, which reduced deer densities to 7 deer km⁻², there was limited recruitment of small trees and declines in basal area of tree species that were sensitive to deer browsing. Our results suggest that recovery from herbivory is a protracted process during which canopy tree regeneration may continue to decline despite a reduction in browsing pressure due to deer culling. Large declines in canopy-tree densities in Carolinian forests may lead to forest size-structures and herbaceous plant communities that resemble rare oak savanna habitat, creating difficult decisions for conservation managers aiming to protect rare and endangered species within native ecosystems. We recommend that managers protect Carolinian forest stands and encourage canopy tree regeneration by increasing seed sources of native trees. While deer control is essential in reducing forest damage, our results highlight the need to explore other forms of active management to expedite otherwise slow increases in tree density.

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1. Introduction

Cervidae (deer) are a family of herbivores that impact forest regeneration throughout temperate regions (Gill, 1992; Coomes et al., 2003; Côté et al., 2004; Dolman and Wäber, 2008; Takatsuki, 2009). Changes in forest structure and composition arising from high deer population densities have important implications for conservation, e.g. reductions in avian biodiversity (Gill and Fuller, 2007). The impacts of native deer populations on forest ecosystems have also increased greatly in Scotland, North America, and Japan over the last century as a result of increasing deer densities associated with anthropogenic activities such as the removal of predators, habitat fragmentation, and milder winters (Clutton-Brock et al., 2004; Côté et al., 2004; Takatsuki, 2009). Although culling

is frequently used as a management tool to reduce deer impacts, little is known about how forest dynamics respond to landscape-level deer herd reductions. Recovery of forest plant communities is a protracted process and may require decades of intensive and sustained reductions in deer densities (Tanentzap et al., 2009). In some instances, simply removing deer may not return a disturbed ecosystem to its previous state (Hobbs and Norton, 1996; Coomes et al., 2003).

White-tailed deer (*Odocoileus virginianus*) are regarded as keystone species in eastern North American forests (Waller and Alverson, 1997). Browsing by deer removes small trees and seedlings from the browse layer (0.5–2.0 m above-ground; Côté et al., 2004), and as gaps are formed in the forest canopy by the death of adult trees, the prevention of plants from colonizing and regenerating within canopy gaps increases understorey light levels (Hynes et al., 2002). Increases in understorey light favour herbaceous plant communities rather than tree saplings, which would have originally persisted (Koh et al., 1996; Ritchie et al., 1998; Tremblay et al., 2006; Tanentzap and Bazely, 2009). Plants that

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establish under greater light availability can prevent the re-establishment of the original deer-preferred saplings (Royo and Carson, 2006), initiating a positive feedback with the potential to shift the ecosystem to an alternate state (Stromayer and Warren, 1997; Augustine et al., 1998). For example, simulations of forest dynamics by Frelich and Lorimer (1985) found that even if deer were removed from systems, changes to forest structure could persist for centuries, with species such as *Tsuga canadensis* failing to recover. Under deer densities of 12–20 deer km⁻², selective browsing alters the competitive abilities of tree seedlings, leading to changes in forest trajectory following canopy disturbance (Eschtruth and Battles, 2008). However, the effects of deer are rarely studied beyond the sapling stage since they require long-term data on the forest overstorey coupled with large-scale manipulations of deer density.

Our objectives were to test whether: (i) prolonged periods of high deer browsing reduce canopy tree regeneration in Carolinian forest after 12-years of monitoring, and (ii) subsequent, intensive deer culling over 16-years restores earlier depleted tree size classes and deer-preferred species. Our study site is the largest remaining tract of the eastern deciduous (Carolinian) forest in Canada, and contains large tracts of Midwestern oak savanna, one of the world's most endangered ecosystems (North American distribution <0.02% of pre-European settlement; Nuzzo, 1986). Both ecosystems are within the Carolinian life zone, which covers 0.25% of the Canadian land area, yet contains at least 124 (ca. 21%) of the country's near-threatened, threatened, and endangered species (OMNR, 2008). To our knowledge, we are the first to monitor long-term patterns of forest tree dynamics in North America in response to deer exclusion using intensive deer herd reductions rather than exclosure experiments (e.g. Horsley et al., 2003). We predicted that during the 12-year period of high deer densities, the forest size-structure would become depleted of small trees in the browse layer. We expected reductions in deer densities over 16-years to increase the abundance of small trees, but that the previous depletion of trees in the forest understorey would lead to reductions in mid- and large-sized trees over time despite deer culling. Reductions in canopy-tree densities may shift Carolinian forests to oak savanna, depending on edaphic conditions and fire history, and therefore, pose a dilemma for conserving multiple rare ecosystems. Lastly, we expected that the forest size-structure in deer exclosures, established during the period of increasing deer populations, would be minimally impacted by browsing and hence resemble browsed-control plots prior to deer population growth.

2. Methods

2.1. Study site

Rondeau Provincial Park, Ontario, Canada (3 254 ha, 42°17'N, 81°52'W) is situated on a cusped sandspit extending out into Lake Erie with a north–south orientation consisting of sand dune ridges and seasonally-flooded sloughs. Species adapted to seasonal submersion in water, such as pumpkin, black, and red ash (*Fraxinus profunda*; *F. nigra*; *F. pennsylvanica*, respectively), and silver maple (*Acer saccharinum*), are primarily located within the sloughs, with the ridges comprised of American beech-hornbeam (*Fagus grandifolia* and *Carpinus caroliniana*, respectively) and sugar maple (*Acer saccharum*) stands with sparse red, black, and white oak (*Quercus rubra*, *Q. velutina*, and *Q. alba*, respectively), tulip tree (*Liriodendron tulipifera*), sassafras (*Sassafras alba*), white ash (*Fraxinus americana*), bitternut hickory (*Carya cordiformis*), black walnut (*Juglans nigra*) and American basswood (*Tilia americana*) (Bartlett, 1958; Figs. S1 and S2). Pine-oak dominated stands (primarily *Pinus strobus* and *Quercus* spp.) occur on the relatively infertile eastern soils (Bartlett, 1958). Overall, shallow and sandy soils that are located above a

high water table limit rooting depth and increase the susceptibility of trees to windthrow (e.g. Woods, 2004), particularly since the park is exposed to strong westerly winds extending over the long and shallow axis of Lake Erie. Catastrophic winds are a major disturbance in northeastern North American forests (Canham and Loucks, 1984; Schulte and Mladenoff, 2005), and the most recent wind disturbance at our site occurred in July, 1998 (Larson and Waldron, 2000). Prior to this, notable windstorms were recorded from 1973 to 1978 (Ontario Parks, unpub. data).

Rondeau has been subject to human disturbance at least since European settlement of the region in the late 1700s. The park itself was established in 1894 for recreation and much of the forest is likely semi-natural secondary growth following logging prior to the park's establishment (Mann and Nelson, 1980). White-tailed deer were intensively hunted, leading to their extirpation by the 1860s (Mann and Nelson, 1980), but were re-introduced in 1901 and are currently the only large mammalian herbivore present and the primary biotic disturbance (Koh et al., 1996). Predators of deer, i.e. wolves (*Canis lupus*), black bears (*Ursus americanus*), and cougars (*Puma concolor*), have been absent in the region for much of the last two centuries, with the exception of at least two coyote (*Canis latrans*) packs that utilize the park. The surrounding landscape of southwestern Ontario is highly fragmented and dominated by agriculture (<5% forest cover; Riley and Mohr, 1994) and human settlements that contain 21% of the Canadian population (Statistics Canada, 2006).

2.2. Deer population size

White-tailed deer population size has been censused regularly in winter at Rondeau by the Ontario Ministry of Natural Resources (OMNR) using drive counts up to 2004 and aerial counts since 2007. The population is relatively closed given the narrow peninsula upon which the park is situated and its large distance from neighbouring forests. Deer densities significantly increased at Rondeau from 27 deer km⁻² in 1974 to 55 deer km⁻² in 1991, after deer gained protected status in Ontario Provincial Parks in 1974 (Mann–Kendall trend test for change in densities between 1974 and 1991: $\tau = 0.52$, $p = 0.024$; Fig. 1). However, deer densities were significantly reduced to 7 deer km⁻² by 2009 following an initial cull that removed 67% of the previous year's population in 1993

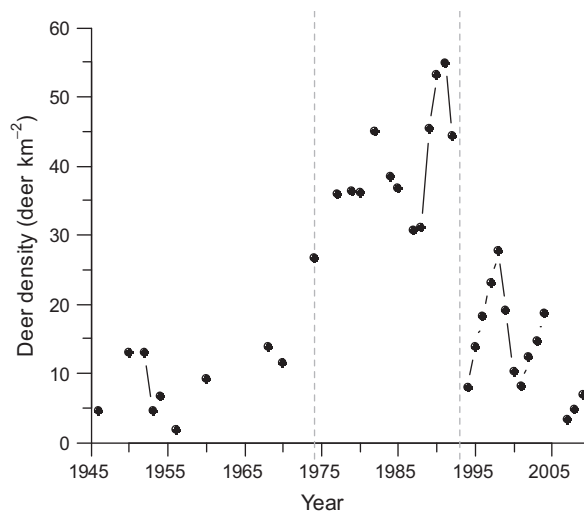


Fig. 1. Estimated annual deer density from 1946–2009 at Rondeau Provincial Park, Ontario, Canada using drive counts (1946–2004) and aerial surveys (2007–2009; Ontario Parks, unpub. data). Gray lines denote cessation and reinstitution of hunting in 1974 and 1993, respectively. Deer populations were not censused in 2005 and 2006.

and annual culls since 1998 (change in densities between 1991 and 2009: $\tau = -0.45$, $p = 0.017$; Fig. 1). All culls have been directed under the supervision of the OMNR, with participation from local First Nations.

2.3. Forest monitoring plots and deer density manipulations

Changes in forest vegetation were surveyed in 19 permanently marked $10 \text{ m} \times 20 \text{ m}$ plots located throughout Rondeau. Plots were independently established on ridges in 1981 within areas with minimal human disturbance (e.g. avoiding campgrounds, roads, and areas where vegetation was cleared). All 19 plots were measured in 1981 and between 1996 and 1997, and 16 plots were measured in 2009. Within each plot, the diameter at breast height (DBH, 1.4 m) was recorded for all live adult trees ($>1.4 \text{ m}$ height) and tree seedlings ($\leq 1.4 \text{ m}$ height) were counted by species in randomly located $1 \text{ m} \times 1 \text{ m}$ subplots ($n = 20$ per plot in 1981, 10 per plot in 1996/97 and 2009). Standing height of all stems was also recorded in 1981 and 2009 using a Suunto clinometer (Suunto Oy, Vantaa, Finland).

We included two large deer exclosures ($50 \text{ m} \times 40 \text{ m}$ and $60 \text{ m} \times 40 \text{ m}$; 4 m high fences) in our study that were established in 1978 and have been regularly maintained (Fig. S3). Locations were not random, but rather positioned for ease of access on medium to dry ridges. We established eight $10 \text{ m} \times 20 \text{ m}$ plots within each exclosure in 2009 and repeated measurements collected in the browsed plots. Additionally, we surveyed the size structure of the exclosures and compared measurements to browsed-control plots in 1993. We randomly established five sampling stations within the large exclosure, and four at the smaller exclosure, and paired an equal number of stations 15 m outside of each exclosure. At each station, a 20 m transect (orientated north–south) was bisected by a 10 m transect and the ends of each transect, and their point of intersection, were marked as sampling points. We selected the closest tree ($>40 \text{ cm}$ in height) to each sampling point and measured the distance from that individual to the closest neighbouring tree and the distance from the second individual to its nearest neighbour. The mean of all the distances recorded at a station ($\bar{D}$, meters, $n = 10$ per station) was used to estimate stem density (stems $\text{ha}^{-1} = \frac{10000}{\bar{D}^2 \times 1.67 \text{ stems}^{-1}}$; Mueller-Dombois and Ellenberg, 1974). We estimated densities separately for adult trees, defined within these plots as stems $>7.5 \text{ cm}$ in DBH, and saplings ($<7.5 \text{ cm}$ diameter at 40-cm height and $>40 \text{ cm}$ in height). A minimum height of 40-cm was intended to differentiate established trees from seedlings and individuals restricted to the ground layer.

We also measured photosynthetically active radiation (PAR; 400–700 μm) at ground-level (0 m) and at a height of 2 m in July 1998 and 1999 using a Licor line quantum sensor (Li-191SA, Licor Biosciences, Lincoln, NE, USA). We established 5–7 plots spaced $>5 \text{ m}$ apart along transects located on either end of one of the four fences comprising each long-term deer exclosure. Values were expressed as a total of available PAR in full sunlight, simultaneously recorded with a Licor point sensor (Li-190SA, Licor Biosciences) in the nearest open area, located $<50 \text{ m}$ away. Measurements were recorded between 11:00 a.m. and 3:00 p.m. on cloudless days and averaged over a period of 15 s to account for sun flecks.

2.4. Statistical analyses

2.4.1. Effects of deer on forest structure and regeneration

We tested whether deer altered forest size structure by comparing the continuous cumulative distribution of tree diameters in forest plots among individual measurement periods. We pooled tree diameters across plots in each year, and compared browsed forest plots between 1981 and 1996, and 1996 and 2009, with two-sam-

ple Kolmogorov–Smirnov tests (*ks.test* in R v2.9; R Development Core Team, 2009). We used the same approach to compare size structure in the browsed forest plots in 1981 and 2009 to exclosure plots. To determine whether changes in size structure were accompanied by changes in basal area, we used paired *t*-tests to compare log-transformed basal area per individual per plot among years using only comparable plots. Lastly, we identified the effects of deer on other measures of forest structure by comparing maximum adult tree height per plot and log-transformed seedling densities between the comparable forest plots measured in 1981 and 2009 using paired *t*-tests.

We determined the effects of deer exclusion on tree density by comparing adult tree and sapling densities per station inside and outside of exclosures in 1993. Although our data conformed to a block design, where exclosure treatments were distributed across the two large exclosure sites, we did not consider exclosure site as a random effect because its influence on model variance would be estimated poorly from only two levels. We fit a generalized linear model with a log-link function in which tree density was a dependent factor, and treatment was nested within block as a fixed factor. The model assumed a negative binomial error structure in order to reduce overdispersion. We determined the significance of exclosure treatment by comparing it to a model with only exclosure site as a fixed factor using a likelihood ratio test with a χ^2 approximation.

We tested whether deer reduced understorey light levels using a linear mixed-model, estimated with unrestricted maximum likelihood (*lmer* function in *lme4* package in R; Bates and Maechler, 2009). Percent PAR was included as a log-transformed dependent variable, and exclosure treatment, height level, sampling year, all two-way interactions and the three-way interaction among these factors, were input as fixed factors nested within exclosure site. Sampling plot was included as a random effect, nested within exclosure site. The full model was reduced by eliminating all single and interaction terms that did not significantly decrease model fit when removed, with likelihood ratio tests ($\alpha = 0.05$).

2.4.2. Changes in forest composition

We used analysis of variance with distance matrices to determine whether the 16 comparable plots changed in composition over time (*adonis* function in *vegan* package in R; Oksanen et al., 2010). We input a Bray–Curtis dissimilarity matrix, calculated using the total basal area of each species per plot, as a dependent variable, with year as a fixed factor and plot as a grouping factor. Basal area was square-root transformed and standardized using a Wisconsin double transformation (Oksanen et al., 2010). Species occurring in only one plot were removed from analyses, and degrees of freedom are not reported since *p*-values were generated from 1000 permutations of the raw data.

We tested whether compositional changes were related to environmental factors by fitting measures of the shade, drought and waterlogging tolerance of individual species (after Niinemets and Valladares, 2006) to their positions in a non-metric multidimensional scaling ordination of plot dissimilarities. We selected the number of ordination axes at which further increases in dimensionality did not appear to substantially reduce stress, which is a measure of the fit between the frequency of occurrence of species and distances between sites in the ordination. We estimated a correlation coefficient for each factor and tested its significance using 1000 random permutations (*envfit* function in *vegan*). We also tested whether compositional changes were related to deer by including a measure of the responses of species to deer exclusion (*V*), estimated from the ratio of individuals $<40 \text{ cm}$ tall inside (N_i) to outside (N_o) of exclosures: $V = \frac{2N_i}{N_i + N_o} - 1$ (Wardle et al., 2001). *V* ranges from -1 (found outside of exclosure but not inside) to 1 (found only within exclosures), and values were estimated using

data collected in June and July 1996 from 10 stations randomly located along each of three 1-km transects. Each station comprised of a 2 and 4 year-old 2 m × 2 m enclosure 5-m on either side of an unfenced 2 m × 2 m control plot (total $n = 60$ enclosures and 30 control plots). Six species present within our permanent forest plots were never recorded within the network of 2 m × 2 m plots: *Populus deltoides*, *P. grandidentata*, *Quercus alba*, *Q. rubra*, *Q. velutina*, and *Ulmus americana*.

3. Results

3.1. Forest size-structure and sapling regeneration

Small trees (<6 cm DBH) were 72% less frequent within plots in 1996 after 19-years of >30 deer km^{-2} than in 1981 when deer densities first began to increase, leading to changes in forest size structure ($D = 0.35$, $p < 0.001$; Fig. 2A and B). The loss of small trees

was accompanied by an increase over time in mean basal area per individual ($\pm$ standard error) across size classes (1981: $147 \pm 17 \text{ cm}^2$; 1996: $299 \pm 51 \text{ cm}^2$; $t_{18} = 4.02$, $p < 0.001$). Although the total abundance of large trees (>20 cm DBH) declined within forest plots by 21% in 1996 compared to 1981, this reduction was much less than that for smaller size classes (<20 cm DBH) of 57% between years (Fig. 2A and B). Reductions in small trees were mirrored by the significantly lower densities of saplings, but not adult trees (>7.5 cm DBH), within browsed plots in 1993 compared to adjacent enclosures ($\chi^2_2 = 13.88$, $p < 0.001$ and $\chi^2_2 = 2.06$, $p = 0.357$, respectively; Fig. 3).

Densities of small trees (<2 cm DBH) increased by 4.5 times between 1996 and 2009 following deer herd reductions, while small to mid-sized trees (2–12 cm DBH) declined by 41% during this period due to previously low abundances of small trees in 1996 that moved upwards through the forest canopy ($D = 0.25$, $p < 0.001$; Fig. 2B and C). Large-sized trees (>40 cm DBH) declined in the forest size structure by 20 trees ha^{-1} between 1996 and 2009, and this was much greater than the reduction in density between 1981 and 1996 of 4 trees ha^{-1} (Fig. 2), suggesting that the 1998 windstorm may have contributed to these reductions. Despite the loss of large trees, mean basal area per individual ($\pm$ standard error) within comparable plots did not change between 1996 and 2009 (1996: $298 \pm 60 \text{ cm}^2$; 2009: $257 \pm 39 \text{ cm}^2$; $t_{15} = 0.92$, $p = 0.371$), in part due to the growth and increased abundance of smaller trees following deer herd reductions. Understorey light availability measured 1-month prior to the 1998 windstorm event was not significantly different from light levels recorded 11 months afterwards (mean $\pm$ standard error: 3.8 ± 0.8 and $3.5 \pm 1.0\%$ PAR, respectively; final model including only the effect of enclosure treatment with and without effect of year: $\chi^2_4 = 4.00$, $p = 0.406$). The lack of a change in understorey PAR suggests that the loss of large trees due to the windstorm did not affect canopy light interception, presumably since these individuals occurred at relatively low densities. The continued decline of large trees did result in lower maximum tree height ($\pm$ standard error) recorded in each plot in 2009 compared to 1981 (1981: $27.8 \pm 1.0 \text{ m}$; 2009: $20.9 \pm 1.2 \text{ m}$; $t_{31} = 22.27$, $p < 0.001$). Seedling densities were also lower in 2009, despite deer culling, than when deer densities began to increase in 1981 ($t_{14} = 4.53$, $p < 0.001$; Fig. 3), but were still greater than in 1996, when no seedlings >50 cm in height were observed.

The densities of small trees within enclosures were more than four times greater than in forest plots, and larger size classes were

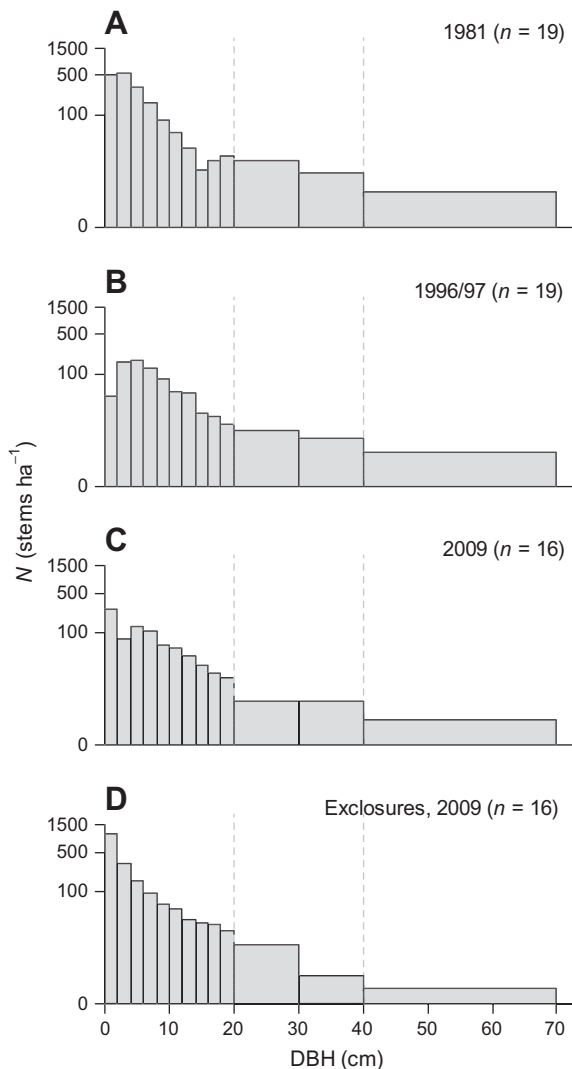


Fig. 2. Diameter distribution of mixed-hardwood forest stands at Rondeau Provincial Park, Ontario. Distributions were obtained by pooling data from 19 plots in (A) 1981 and (B) 1996/97, (C) 16 plots in 2009, and (D) 16 enclosure plots in 2009. Individual stems (>1.4 m height) were assigned into 2-cm wide diameter-size classes for trees ≤ 20 cm DBH, 10-cm wide size classes for trees with DBH >20 cm but ≤ 40 cm, and 30-cm wide size class for trees with DBH >40 cm. The total numbers of stems in each size class are plotted in (A–D) on a log-scale, with values standardized per unit cm to allow for comparison across size class divisions (i.e. 2, 10, and 30 cm).

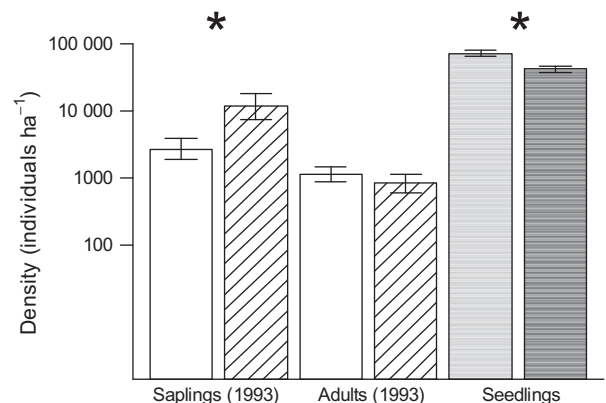


Fig. 3. Mean densities $\pm$ standard error of saplings and adults in browsed-control plots (open bars) and enclosures (hatched bars) in 1993 (for each, $n = 9$), and seedlings in permanent forest plots in 1981 (light gray) and 2009 (dark gray). Saplings were <7.5 cm diameter at 40-cm height and >40 cm in height, adults were >7.5 cm DBH, and seedlings were ≤ 1.4 m height irrespective of DBH. *Denotes significant differences ($\alpha = 0.05$) between saplings in controls and enclosures, and seedlings in 1981 and 2009 (see text for details).

more abundant within exclosures in 2009 ($D = 0.33$, $p < 0.001$; Fig. 2C and D). The size-structure of the exclosures also differed from that within forest-plots in 1981, with greater densities of small and large (>20 cm DBH) trees within exclosures ($D = 0.36$, $p < 0.001$; Fig. 2). Light levels were significantly greater outside of the exclosures due to lower tree densities (mean $\pm$ standard error: 3.2 ± 1.0 and $4.1 \pm 0.8\%$ PAR in exclosures and control plots, respectively; final model with only effect of exclosure treatment nested within exclosure site versus model with only exclosure site: $\chi^2_2 = 18.74$, $p < 0.001$).

3.2. Forest community composition

The composition of basal area within plots significantly differed over time ($F_{2,47} = 0.86$, $p = 0.002$). Following deer herd reductions (1996–2009), virtually all plots increased along axis 1 and 2 of the ordination (14/16 and 15/16 plots, respectively), which were positively associated with a plot-level basal area comprised of species that increased in abundance in the presence of deer (correlation between V index and position of plots in ordination space: $r^2 = 0.63$, $p = 0.029$; $n = 15$). Species that were presumably sensitive to deer browsing, and therefore, increased in abundance only within deer exclosures ($V > 0$), comprised less of the total basal area of plots in 2009 than in 1996 (Fig. 4). There were no consistent

temporal changes in within-plot composition along any axis between 1981 and 1996 and composition was unrelated to the water, shade, and drought tolerance of species across time periods (for all, $r^2 < 0.29$, $p > 0.200$).

4. Discussion

4.1. A legacy of deer herd protection

Our study is among the first to demonstrate continued declines in canopy trees despite some understorey regeneration during intensive deer herd reductions. This finding suggests that culling alone may only reduce rather than eliminate the negative impacts of high deer densities (>10 deer km^{-2}) on tree density and composition observed at our site and in many other temperate forests (Putman et al., 1989; Veblen et al., 1989; Husheer et al., 2003; Mountford and Peterken, 2003). Reductions in small trees associated with high deer densities can have cascading effects into larger size classes over time, leading to continual declines in canopy trees without any regeneration even when browse pressures are removed. Declines in canopy tree density that enlarge canopy gaps or reduce canopy tree height may further increase wind speeds (Tanentzap et al., 2007; Vygodskaya et al., 2007; Panferov and Sogachev, 2008), and thus act as a positive feedback to further increase the susceptibility of large-sized trees to windthrow (Canham et al., 2001). Similarly, declines in tree density that reduce lateral rooting mass, and hence soil stability, may further increase the susceptibility of trees to windthrow (Zhou et al., 1997). Reductions in canopy-tree densities deplete seed sources and may strongly limit the potential for increased seedling and sapling densities following deer culling.

We observed some recruitment of small trees within forest plots by 2009, suggesting that recovery from herbivory may be occurring but is an extremely protracted process. In New Zealand *Nothofagus* forests, reductions in deer densities over 44 years similarly led to small increases in the densities of deer-preferred tree seedlings and saplings (Tanentzap et al., 2009). Hardwood species may require >10 years of deer exclusion in order for seedling recruitment to increase and in part, this “legacy” of deer browsing may persist long after browse pressures are reduced because of depletion in seed sources (Long et al., 2007). The establishment of species that respond positively to deer browsing may also result in shifts in community composition that persist once deer are removed; e.g. browse-tolerant and herbivore-avoided graminoids and ferns that form thick monocultures under high deer densities more effectively compete for light than tree seedlings and limit canopy tree regeneration (Royo and Carson, 2006). Once seedlings are able to establish, long periods may still be required to escape the browse layer, e.g. late successional conifers require >30 years of protection from deer browsing in order to reach higher size classes (Anderson and Katz, 1993; Cornett et al., 2000). These responses may arise because of highly non-linear relationships between deer densities and seedling regeneration (Tremblay et al., 2006; Gill and Morgan, 2010), and the relatively slow growth of long-lived canopy tree species (Cornett et al., 2000). Where forage is relatively uncommon, reductions in deer densities may not be accompanied by linear increases in seedling densities because deer may simply broaden their diets to include less-preferred forage (e.g. leaf litter) rather than avoid consuming live forage (Coomes et al., 2003). Deer removal can also increase the abundance of small mammals, reducing tree regeneration through seed predation in the absence of deer (Smit et al., 2001). Intense disturbances (e.g. severe windstorms) may be required to facilitate tree regeneration at some sites, and therefore, simply excluding deer browsing may be inadequate to increase tree densities in the short

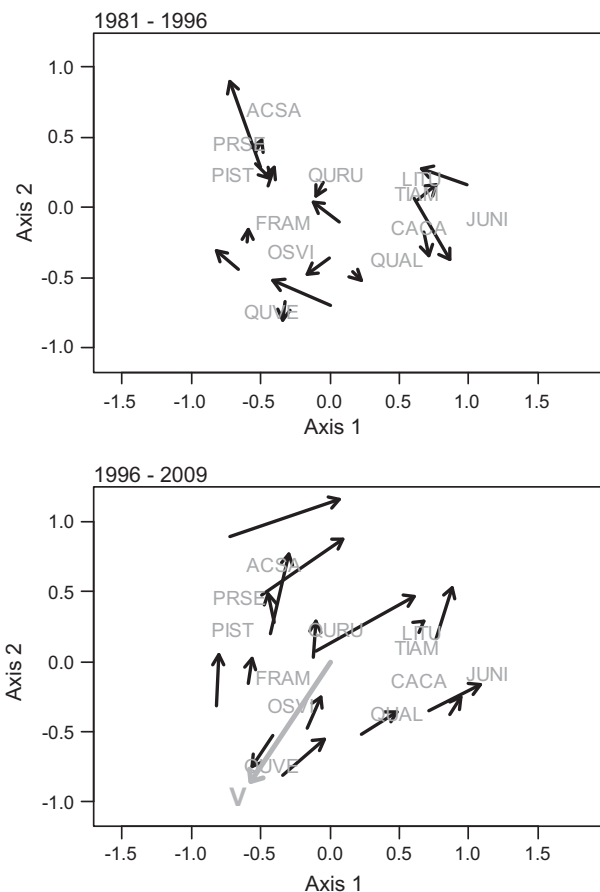


Fig. 4. Non-metric multidimensional scaling ordination of basal area with Bray-Curtis distance measures for 16 permanent forest plots at Rondeau Provincial Park, Ontario (stress = 13.97). Black arrows indicate direction of change in plots from 1981 to 1996 and 1996 to 2009. Twelve most common species are plotted: ACSC = *Acer saccharum*; CACA = *Carpinus caroliniana*; FRAM = *Fraxinus americana*; JUNI = *Juglans nigra*; LITU = *Liriodendron tulipifera*; OSVI = *Ostrya virginiana*; PIST = *Pinus strobus*; PRSE = *Prunus serotina*; QUAL = *Quercus alba*; QURU = *Q. rubra*; QUVE = *Q. velutina*; and TIAM = *Tilia americana*. We also plotted the vector of species responses to deer exclusion (V , dark gray).

term. Disturbances increase understorey light levels and/or mineral nutrient availability, stimulating seed germination and tree growth, but can also facilitate invasions by non-indigenous trees that outcompete native species (Carter and Fredericksen, 2007). In some instances, disturbances create important refugia from browsing (e.g. tip-up mounds) that increase seedling establishment (Long et al., 1998).

Successional patterns and wind disturbance may be important in influencing our observed patterns of forest size structure. However, during the period of increased basal area and declines in stem density within our plots (1981–1996), the effects of deer were clearly evident from the greater densities of small trees within exclosures. Our results therefore suggest that deer herbivory may accelerate declines in stem density that would otherwise occur in response to resource competition during succession. In Japanese temperate deciduous forests, declines in understorey tree density due to bark stripping and browsing by sika deer (*Cervus nippon*) increase the accumulation of biomass, similar to artificial thinning of managed forest stands (Miyaki and Kaji, 2009). We also report increased tree recruitment during the period of deer herd reductions (1996–2009). These results provide further evidence for deer rather than succession or wind-induced canopy gaps as the principal driver of forest regeneration at our site, particularly since understorey light levels did not increase after the 1998 windstorm. Preferential browsing of fast-growing shade intolerant species by deer (Long et al., 2007; Krueger et al., 2009) may limit stem recruitment following catastrophic wind events (Dyer and Baird, 1997), and low levels of forest regeneration can drastically alter forest trajectories after canopy gap creation (Peterson and Pickett, 1995; Eschtruth and Battles, 2008).

4.2. Conserving rare habitats maintained by competing processes

A lack of canopy tree regeneration has begun to shift the forest at our site towards an open-canopy ecosystem, the most phytosociologically-similar of which is oak savanna (Curtis, 1959; Tester, 1989; Peterson and Reich, 2001; Brudvig and Asbjørnsen, 2007). Key indicator species of oak savanna understories, such as *Carex pennsylvanica*, have also increased in abundance within browsed plots compared to deer exclosures, in addition to ruderal species characteristic of open-canopy sites (Koh et al., 1996). However, the persistence of oak savanna requires natural fire regimes (Tester, 1989; Peterson and Reich, 2001; Brudvig and Asbjørnsen, 2007), which are severely suppressed within the highly populated region of northeastern North America (Nowacki and Abrams, 2008), including our study site. Since oak savanna is a globally rare ecosystem, our study raises the question of whether managers should capitalize on reductions in tree densities to manage low-density open-canopy stands as oak savanna through prescribed burning and high deer densities, particularly where low levels of key savanna indicator species, e.g. *Quercus velutina* or *Carex pennsylvanica*, already persist. We suggest that such an approach is flawed because the transition from Carolinian forest into oak savanna is not a retrogressive process (Curtis, 1959; Rogers et al., 2008), and would require the promotion of an ecosystem without any guarantee of success, as opposed to preservation of existing but degraded Carolinian forest. Furthermore, the most efficient use of limited resources for conservation decisions is not to oppose the self-organization of ecosystems; i.e. open-grasslands develop into forests in the absence of disturbance (Holdaway et al., 2010). Although disturbances by large herbivores may have maintained open-canopy woody-grassland mosaics in Europe (Kirby, 2004; Birks, 2005), these processes require the presence of fire in North America (Ritchie et al., 1998).

We recommend that managers continue to protect Carolinian forest stands. Management strategies to restore and expand forest

stands should focus on increasing seed sources of native trees to mitigate declines in canopy-tree densities, in addition to maintaining deer densities at their present levels (<7 deer km^{-2}). Protecting established saplings and planting species within small deer-exclosures might further accelerate recovery and large-scale plantings may lessen windthrow by reducing wind speeds (Tanentzap et al., 2007). The restoration of ecological processes, e.g. predator re-introduction, may be equally desirable. For example, the re-introduction of wolves in Yellowstone National Park reduced relatively high densities (>10 animals km^{-2}) of elk (*Cervus elaphus*), allowing heights of aspen trees (*Populus tremuloides*) to increase for the first time since wolves were originally extirpated (Ripple and Beschta, 2007). Similar trophic cascades elsewhere emphasize the strong controls that large carnivores exert on tree regeneration through herbivore predation (Hebblewhite et al., 2005; Ripple and Beschta, 2008). However, predator re-introduction may be impractical within densely populated regions. Overall, our results highlight that forest recovery following high deer populations is a protracted process, requiring decades of commitment from managers despite limited immediate success.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.biocon.2010.09.015.

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